

A Mechanical Route to G : Matching Field Stress–Energy to the Einstein–Hilbert Coupling

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Abstract

We develop a mechanical route to Newton’s gravitational coupling G by relating the macroscopic field stress–energy to the geometric stiffness of the Einstein–Hilbert (EH) action. In the long–wavelength regime of a homogeneous, barotropic background, the bulk modulus satisfies $K = \rho_m c_s^2$; identifying the signal speed with c yields $K = \rho_m c^2$. We then show that an EH term $\int \sqrt{-g} R$ arises with a calculable coefficient in the one–loop effective action, so that matching the geometric stiffness $c^4/(8\pi G)$ to the mechanical stiffness gives $G = c^2/(8\pi \rho_m)$. Independently, we verify the observational identity $G = \Lambda c^4/(8\pi u_\nu)$, where Λ is the cosmological constant and u_ν is the corresponding vacuum stress–energy density, providing a parameter–free cross–check.

To connect with microphysics and phenomenology, we outline three realizations: (A) an elastic curvature response consistent with thermodynamic and induced–gravity perspectives; (B) a superfluid dark–matter realization in which phonon excitations couple to baryons and exert a literal force; and (C) a single–field EFT of dark energy with explicit matter couplings that generate an effective force and $G_{\text{eff}}(t, k)$. Our baseline adopts (A) (no extra force) and recovers Λ CDM at linear order; (B) and (C) are presented as optional model ingredients with distinct, testable signatures.

1 Introduction

Analogue–gravity studies show that hydrodynamic media can reproduce relativistic kinematics via an effective metric,^{1–3} suggesting a conservative route to relate gravitational dynamics to mechanical properties. In this work, we develop a mechanical route to G by modeling the background as a barotropic medium. Identifying the signal speed with c fixes the field pressure to $K = \rho_m c^2$, which we match to the geometric stiffness multiplying curvature in the Einstein–Hilbert (EH) action.^{4,5}

2 Assumptions and Regime of Validity

We assume: (i) long–wavelength hydrodynamics ($k\xi \ll 1$); (ii) homogeneous, barotropic background with mass density ρ_m ; (iii) phase coherence; (iv) identification of acoustic and electromagnetic null cones in the IR ($c_s = c$).

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3 Field Pressure: Three Realizations and Couplings

We use “field pressure” as shorthand for the macroscopic stress–energy that sets a response scale for spacetime. The way this response acts on matter depends on microphysics and couplings. We distinguish three realizations:

(A) Elastic curvature response (baseline). With minimal coupling and a cosmological constant background, the field pressure fixes the medium’s elastic response; matter follows geodesics of the curved metric. The “push back” is spacetime curvature—an equation-of-state relation—not a direct pressure force on baryons. ^{13–15}

(B) Superfluid dark matter with phonon force (optional). If the field is a dark-matter superfluid in galaxies, the phonon mode θ can couple to baryons, e.g. $\mathcal{L}_{\text{int}} = \alpha \theta \rho_b$, so baryons feel an additional force $\mathbf{a}_{\text{phonon}} = -\alpha \nabla \theta$ on top of geodesic motion. Here pressure/stress gradients transmit a literal force in the superfluid phase, while cosmology remains CDM-like. ^{16,17}

(C) Single-field EFT of dark energy with explicit couplings (optional). In a general single-field EFT, the background field supplies the pressure/energy density, and matter may couple conformally/disformally to the scalar sector. At the perturbative level this produces an effective force on baryons and a scale- and time-dependent $G_{\text{eff}}(t, k)$, subject to stability and observational constraints; setting extra quadratic operators to zero reproduces Λ CDM at linear order. ¹⁸

Constraint summary. Any direct baryon–vacuum interaction producing a fifth force is excluded at the level $|\alpha| \lesssim 10^{-13}$ (and below) by precision equivalence–principle tests unless it is screened or emergent; therefore, in the baseline scenario baryons couple to vacuum stress–energy only through spacetime curvature, while literal forces arise only in the optional superfluid or EFT realizations. ^{13,16,18–20}

4 Hydrodynamic Framework

At the microscopic level, we take a complex scalar condensate described by the Gross–Pitaevskii (GP) equation: ^{6,7}

$$i\hbar \partial_t \Psi = \left[-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{ext}} + g|\Psi|^2 \right] \Psi. \quad (1)$$

Writing $\Psi = \sqrt{n} e^{i\theta}$ gives hydrodynamics for number density n and velocity $\mathbf{v} = (\hbar/m)\nabla\theta$. Linear perturbations propagate on an acoustic metric $ds^2 \propto -c_s^2 dt^2 + d\mathbf{x}^2$.

5 Mechanical Stiffness and Geometric Coupling

5.1 Bulk Modulus from the Equation of State

For the GP fluid, the pressure is $P = \frac{1}{2}gn^2$. We define $\rho_m \equiv mn$ as the mass density of the condensate (where m is the constituent mass and n is the number density). The sound speed is determined by the derivative of pressure with respect to the mass density: ⁸

$$c_s^2 = \frac{dP}{d\rho_m} = \frac{d(\frac{1}{2}gn^2)}{d(mn)} = \frac{gn}{m}. \quad (2)$$

The bulk modulus (mechanical stiffness) is $\beta \equiv \rho_m \frac{dP}{d\rho_m}$. Setting $c_s = c$ yields:

$$K \equiv \beta = \rho_m c^2. \quad (3)$$

5.2 Matching to Einstein’s Coupling

The EH action has the prefactor $c^4/(16\pi G)$, corresponding to a geometric stiffness $c^4/(8\pi G)$. Equating mechanical and geometric stiffnesses gives:

$$\rho_m c^2 = \frac{c^4}{8\pi G} \implies G = \frac{c^2}{8\pi \rho_m}. \quad (4)$$

5.3 Verification and Significance

The identity

$$G = \frac{\Lambda c^4}{8\pi u_\nu} \quad (5)$$

follows from the standard equivalence between the cosmological constant term $\Lambda g_{\mu\nu}$ and a Lorentz-invariant vacuum stress–energy $T_{\mu\nu}^{(\Lambda)} = -u_\nu g_{\mu\nu}$ with $u_\nu = \rho_\Lambda c^2 = \Lambda c^4/(8\pi G)$. Using observational values for Λ and u_ν yields $G \simeq 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, a parameter-free cross-check. In our constitutive interpretation, the background vacuum stress–energy (associated with Λ) sets the elastic response scale of spacetime; G is the corresponding modulus. Realizations (B) and (C) introduce explicit matter couplings through which pressure/stress gradients can exert a literal force on baryons; our baseline analysis adopts (A) with no extra force and hence reduces to Λ CDM at linear order. ^{13,16,18}

Interpretive consistency. Our viewpoint is conceptually consistent with: (i) Jacobson’s reading of the Einstein equations as an *equation of state*; (ii) Sakharov’s induced-gravity picture in which the Einstein–Hilbert term encodes a *metric elasticity* generated by vacuum fluctuations; and (iii) Padmanabhan’s paradigm of *gravity as spacetime elasticity*. Accordingly, we take the background vacuum stress–energy to set the *macroscopic stiffness* of the gravitational medium. Newton’s constant G then appears as a *constitutive parameter* (effective modulus) quantifying spacetime’s resistance to curvature; matter curves spacetime, and the elastic response is fixed by this modulus. ^{13–15}

5.4 Novel Correlation and Derivation

While the Gross–Pitaevskii equation describes a condensed medium and the heat-kernel expansion organizes the gravitational effective action, these frameworks are usually treated separately. Our contribution is the *stiffness matching* condition: in the IR (long-wavelength) regime, the mechanical bulk modulus is matched to the EH geometric stiffness, turning G from an empirical constant into a constitutive parameter fixed by the background state. This matching is compatible with an elastic/thermodynamic reading of spacetime and admits microphysical realizations with and without explicit forces on baryons.

6 Dimensional Consistency and Physical Interpretation

To ensure the mathematical validity of the proposed mechanical route, we perform a dimensional analysis of the identity $G = \Lambda c^4/(8\pi u_\nu)$. As shown in Table 1, the units of the derived coupling perfectly match the standard dimensions of Newton’s gravitational constant.

Physical Significance of the Result

The dimensional alignment supports the following physical interpretations:

Table 1: Dimensional Analysis of the Mechanical Route to G

Quantity	Symbol	SI Units	Dimensions
Gravitational Constant	G	$\text{m}^3 \text{kg}^{-1} \text{s}^{-2}$	$[L]^3 [M]^{-1} [T]^{-2}$
Speed of Light	c	m s^{-1}	$[L] [T]^{-1}$
Vacuum Energy Density	u_ν	$\text{kg m}^{-1} \text{s}^{-2} \text{ (Pa)}$	$[M] [L]^{-1} [T]^{-2}$
Cosmological Constant	Λ	m^{-2}	$[L]^{-2}$
Condensate Density	ρ_m	kg m^{-3}	$[M] [L]^{-3}$

Table 2: Algebraic Breakdown of Dimensional Consistency

Component	Dimensional Operation
Numerator ($[\Lambda] \times [c]^4$)	$[L]^{-2} \times ([L][T]^{-1})^4 = [L]^2 [T]^{-4}$
Denominator ($[u_\nu]$)	$[M][L]^{-1}[T]^{-2}$
Final Ratio ($\frac{\text{Numerator}}{\text{Denominator}}$)	$\frac{[L]^2 [T]^{-4}}{[M][L]^{-1}[T]^{-2}} = [L]^3 [M]^{-1} [T]^{-2}$
Identity Check	$\equiv [G]$

- **Energy density as field pressure:** The vacuum energy density u_ν (Pa) sets the macroscopic scale of the medium’s resistance; in the baseline (A) it fixes the elastic curvature response, while in (B) and (C) pressure/stress gradients can also exert literal forces when couplings are present.
- **Curvature as deformation:** The cosmological constant Λ (inverse length squared) characterizes a uniform curvature scale of spacetime.
- **Gravity as an emergent stiffness:** The gravitational constant G is the effective modulus linking stress–energy to curvature in the IR description.

7 Phenomenological Outlook (Optional Realizations)

The superfluid realization (B) predicts halo–scale signatures (e.g., phonon–mediated forces, cores, and phase–dependent dynamics), while the EFT realization (C) organizes linear–regime deviations via $G_{\text{eff}}(t, k)$ under stability and observational constraints. Our baseline (A) introduces no extra force and reproduces standard Λ CDM growth and lensing at linear order. A dedicated exploration of (B)/(C) against data is left for future work.

8 Conclusion

We have shown that Newton’s gravitational constant G can be interpreted as a constitutive parameter by matching the mechanical bulk modulus of a barotropic medium to the geometric stiffness of the Einstein–Hilbert action. An observational identity using the cosmological constant and vacuum stress–energy density provides a parameter–free cross–check of the numerical value. The framework is consistent with an elastic/thermodynamic reading of spacetime and admits realizations with and without explicit forces on baryons. Our baseline recovers Λ CDM at linear order; optional superfluid and EFT realizations offer distinct, testable signatures.

Declarations

AI Transparency Statement

The fundamental theoretical framework, the correlation matching mechanical bulk modulus to geometric stiffness, and the conceptual interpretation of G are the original work of the author. Generative AI tools were used for LaTeX formatting, bibliography organization, and prose polishing.

A One-loop derivation of the EH term

We integrate out field fluctuations. The divergent part of the one-loop effective action Γ_1 is given by the Schwinger proper-time representation:^{9,10}

$$\Gamma_1 = -\frac{1}{2(4\pi)^2} \int d^4x \sqrt{-g} \int_{1/\Lambda_{UV}^2}^{\infty} \frac{ds}{s^3} e^{-sM_{\text{eff}}^2} [a_0 + a_1 s + a_2 s^2]. \quad (6)$$

The term $a_1 = (\frac{1}{6} - \xi)R$ induces the EH action:

$$S_{\text{ind}} \supset \frac{(\frac{1}{6} - \xi)\Lambda_{UV}^2}{32\pi^2} \int d^4x \sqrt{-g} R. \quad (7)$$

Matching this to $S_{\text{EH}} = \frac{c^4}{16\pi G} \int \sqrt{-g} R$ anchors G to the medium’s cutoff Λ_{UV} in the induced-gravity picture.

B Stress Test: Objections and Responses (FAQ)

Q1: Is this just GR with Λ ? Where is the novelty?

A: The field equations are standard. The novelty is interpretive and structural: we treat the background vacuum stress–energy as primary and read G as a macroscopic modulus, aligned with thermodynamic/induced–gravity perspectives.^{13–15}

Q2: Does “field pressure” push directly on matter?

A: In the baseline (A), no: the response is elastic curvature (geodesic motion). In realizations (B) and (C), explicit couplings let pressure/stress gradients exert a literal force (phonon-mediated in a superfluid; scalar–matter couplings in EFT).^{16,18}

Q3: Does this imply a varying G ?

A: Not in the baseline, which assumes a homogeneous, quasi-constant background ($w \simeq -1$). Generalizations with $G_{\text{eff}}(t, k)$ fit naturally into EFT-of-DE and are constrained by growth/lensing; we leave quantitative fits to future work.¹⁸

C Illustrative Extension: Evolution Across Epochs (Hypothetical)

The mechanical interpretation suggests a possible relation between G and the medium density ρ_m . If ρ_m were to track cosmic evolution, one could obtain a varying $G \propto \rho_m^{-1}$. This is a *hypothetical* extension subject to stringent constraints (BBN, CMB, local tests); we do not claim such variations are realized in nature. The table below illustrates the scaling logic only.

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Table 3: Illustrative scaling in a hypothetical tracking regime

Epoch	Redshift (z)	Rel. Density ρ_m/ρ_0	Rel. Coupling G/G_0
Radiation Dom.	> 3000	$\sim 10^{10}$	$\sim 10^{-10}$
Matter Dom.	~ 1	8	0.125
Present Day	0	1	1

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